

M21C Long-Term Trends and Observing-System Breakpoints

Open-loop background behavior, data-assimilation impacts, and observing-system transitions

August 13, 2026

Executive Summary

This report examines whether the evolving GEOS LDAS data-assimilation (DA) system introduced long-term trends or abrupt discontinuities into precipitation, snow, and soil-moisture states during June 2000-May 2024. It compares the matched open-loop (OL v2) and DA (DA v3) simulations, tests the trend in the strictly paired DA – OL series, evaluates changes at known observing-system transitions (P1-P9), and independently searches for changepoints.

The main conclusion is reassuring: **DA generally preserves the long-term state behavior of OL rather than imposing a new global drift.** Precipitation trends are nearly identical in OL and DA. Snow mass and snow depth show effectively no long-term trend in either experiment. Snow-covered fraction declines in the area-weighted seasonal-snow-domain mean, but it declines at almost exactly the same rate in OL and DA, while DA – OL has no trend. Surface soil-moisture effects are limited.

Root-zone soil moisture (RZMC) is the qualified exception. DA expands the area with significant regional trends, and significant DA – OL trends are mostly positive, particularly in dry and transitional regions. However, OL, DA, and DA – OL still have no significant global land-mean RZMC trend. DA therefore modifies the **regional pattern** of root-zone trends without creating a resolved global secular trend.

The clearest observing-system discontinuity occurs in **April 2015**, when SMAP brightness-temperature assimilation begins. Known-date inference and independent changepoint detection converge exactly on this date across RZMC DA – OL, surface and root-zone correction diagnostics, and total soil-water correction activity. This is strong controlled evidence for a system transition, although it is not proof that SMAP alone caused every contemporaneous change.

Questions and Scope

The analysis addresses four questions:

1. Does the OL simulation contain long-term precipitation, snow, or soil-moisture trends?
2. Does DA preserve, amplify, or oppose those background trends?
3. Do known observing-system transitions produce detectable level or slope changes?
4. Do independently detected changepoints agree with those known dates?

The record contains 288 monthly values from June 2000 through May 2024 on the 112,573-tile M36 land grid. Snow-state analyses use a Northern Hemisphere seasonal-snow mask containing 48,067 tiles. Maps extend to 60 degrees S and use the Robinson projection. All reported domain means are weighted by M36 tile area.

Two spatial scales must be kept separate:

- **Whole-record trend maps** are calculated independently at each tile.

- **Known-transition and independent-changepoint analyses** use one area-weighted monthly domain-mean series for each variable and mask.

Data and Statistical Design

OL and DA values are restricted to the same finite monthly samples before any trend or difference is calculated. The primary DA-impact estimand is the trend of the paired monthly DA – OL series, not the difference between two independently estimated slopes.

Tile trends use an exact Theil-Sen slope after a trend-preserving calendar-month adjustment. Significance uses a modified Mann-Kendall test with an autocorrelation variance correction, followed by Benjamini-Hochberg false discovery rate (BH-FDR) control at 0.05 across every complete spatial field. OL, DA, and DA – OL each retain their own full-field FDR family.

Area-weighted OL, DA, and DA – OL domain trends for six state variables form one additional 18-test BH-FDR family. Known-date transition inference uses a P1-P9 segmented interrupted time-series model with Prais-Winsten AR(1) correction and fitted-AR(1) bootstrap intervals. P7 is treated as level-only because its 15-month duration cannot support an independent slope estimate.

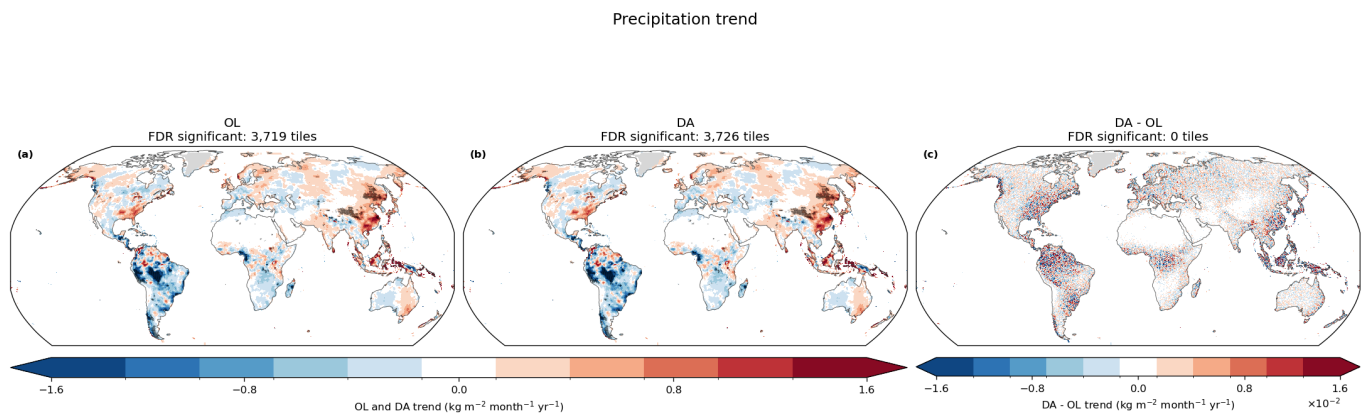
Independent changepoints use two PELT formulations across a penalty grid. A break is accepted only when it is present under the primary AR(1)+linear model, stable across penalties, and reproduced by an independently prewhitened linear method. This conservative consensus is designed to resolve abrupt structural changes; validation shows that it has little power to localize gradual slope hinges.

Whole-Record Trends

Precipitation: a common-forcing control

OL precipitation has significant trends at 3,719 tiles (3.30%), and DA at 3,726 tiles (3.31%). Of these, 3,603 tiles overlap, every overlapping trend has the same sign, and the complete OL/DA slope correlation is 0.9998. No DA – OL precipitation tile survives FDR, and none of the area-weighted OL, DA, or DA – OL precipitation trends is significant.

Interpretation: regional precipitation trends exist in the forcing, but DA does not measurably alter them.

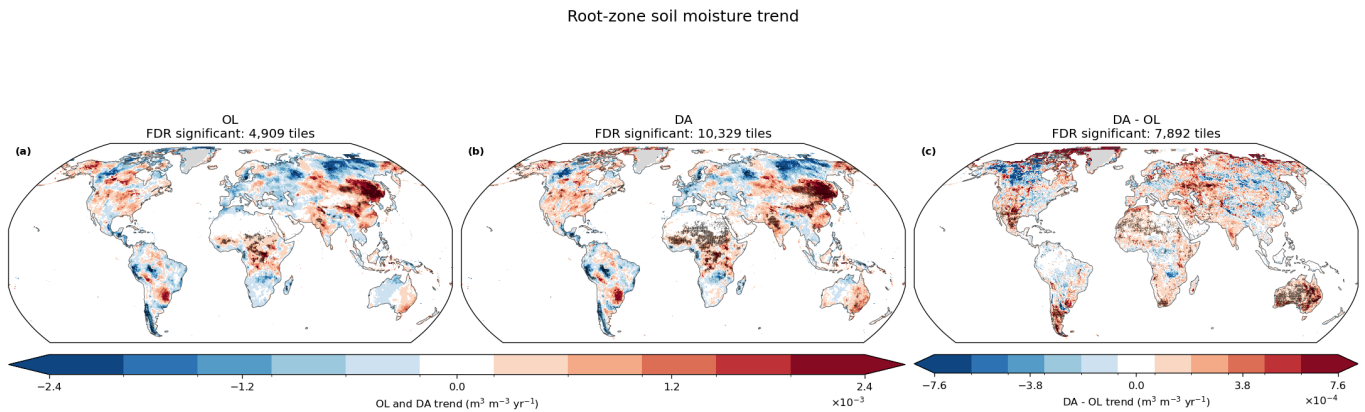


Soil moisture: limited surface effects, clearer regional root-zone effects

Surface soil moisture (SFMC) contains a modest background trend pattern: 6,992 OL tiles (6.21%) and 8,966 DA tiles (7.96%) are significant. Only 1,412 tiles (1.25%) have a significant DA – OL trend, and the area-weighted SFMC trends are not significant. DA modifies SFMC trends locally but does not create a global mean drift.

RZMC is more responsive. Significant coverage increases from 4,909 OL tiles (4.36%) to 10,329 DA tiles (9.18%). The DA – OL trend is significant at 7,892 tiles (7.01%); 7,267 of these are positive and 625 negative. Positive effects are especially extensive in Australia, North Africa and the Middle East, southern Africa, and parts of the Americas and northern Eurasia. Nevertheless, the area-weighted OL, DA, and DA – OL RZMC trends are all nonsignificant.

Interpretation: DA increasingly shifts the root zone wetter relative to OL in particular regions, but it does not introduce a significant global land-mean RZMC trend.



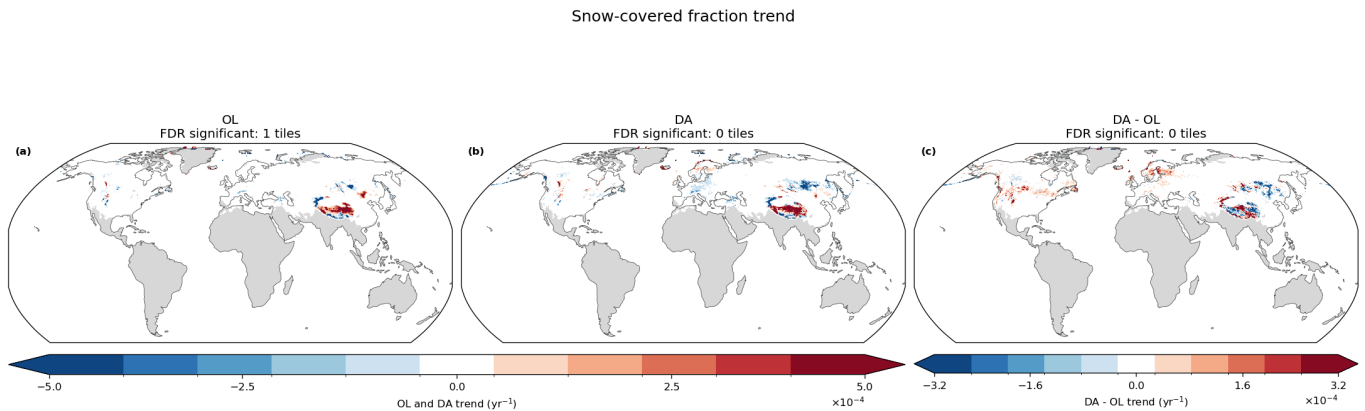
Snow: no DA-induced trend

Snow mass and snow depth are effectively null at the tile scale. Snow mass has 12 significant OL tiles and 5 DA tiles out of 48,067; snow depth has 7 and 6, respectively. No DA – OL tile survives FDR for either variable, and their area-weighted domain trends are nonsignificant.

Snow-covered fraction provides a more subtle result. Its area-weighted seasonal-snow-domain mean declines significantly in both experiments:

Series	Sen slope (yr-1)	Autocorrelation-adjusted 95% interval	BH q
OL	-0.000554	-0.000911 to -0.000198	0.0061
DA	-0.000549	-0.000856 to -0.000233	0.0016
DA - OL	-0.000006	-0.000073 to +0.000065	0.9934

The OL and DA slopes imply an approximately 0.013 reduction in domain-mean snow-covered fraction over 24 years. Their near equality, together with the null DA – OL trend, shows that DA preserves rather than creates the decline. The domain mean can resolve a small coherent change even though individual tile effects are too weak to survive correction across 48,067 spatial tests.



Observing-System Transitions

April 2015 / SMAP is the clearest transition

P6 begins in April 2015, when SMAP brightness-temperature assimilation enters the observing system. Nine primary series have FDR-significant known-date level changes at P6, including:

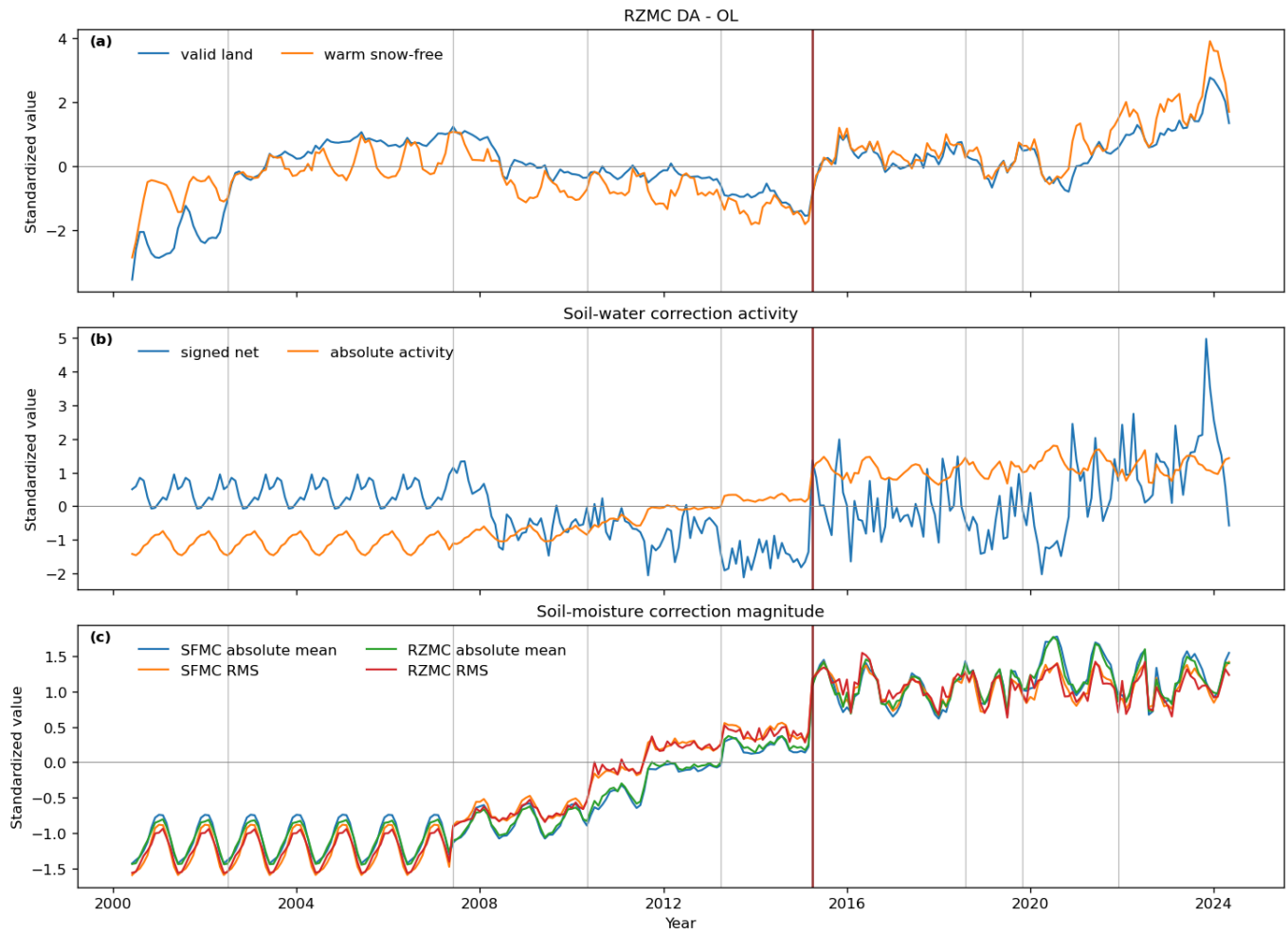
Primary estimand	P6 level change	95% bootstrap interval
RZMC DA – OL, valid land	+0.001021 m3 m-3	+0.000488 to +0.001558
RZMC DA – OL, warm snow-free	+0.001267 m3 m-3	+0.000436 to +0.002037
Soil-water absolute activity	+10.612 kg m-2	+7.056 to +14.715
Soil-water signed net correction	+1.530 kg m-2	+0.589 to +2.467
SFMC increment RMS	+0.001107 m3 m-3	+0.000677 to +0.001505
RZMC increment RMS	+0.000120 m3 m-3	+0.000077 to +0.000162

The independent method finds ten primary series breaking exactly in April 2015. Nine are also significant in the known-date analysis; the additional series is signed SFMC increment mean. No paired OL or DA control series has an accepted independent break. The result is best stated as follows:

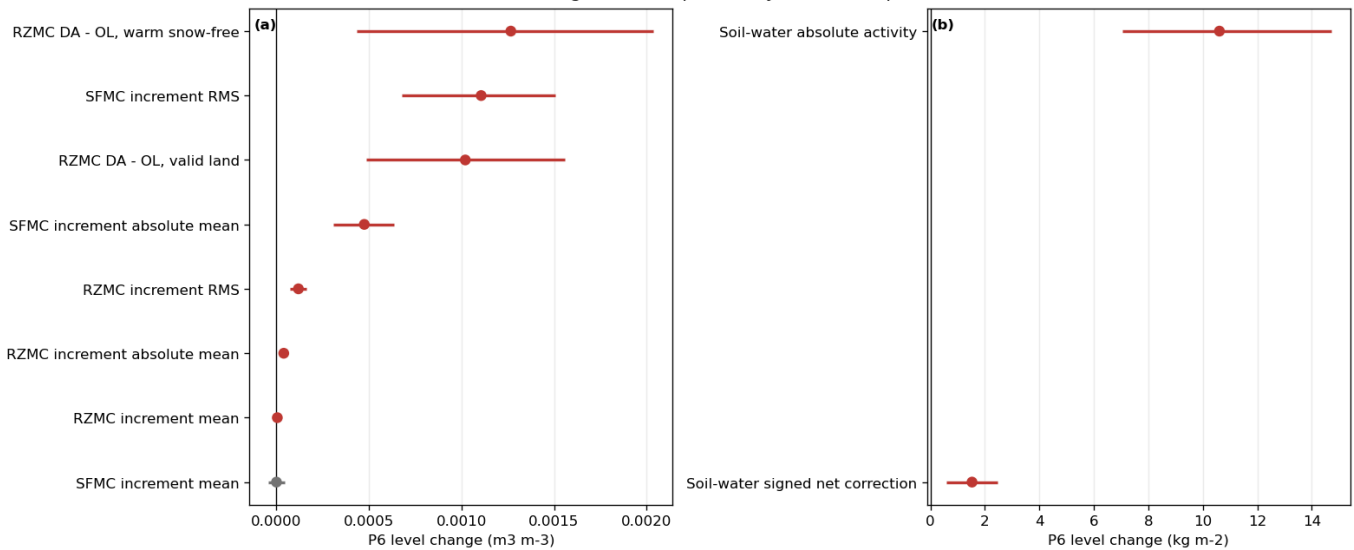
The introduction of SMAP coincides with an abrupt increase in soil-moisture correction activity and a small positive shift in the DA impact on root-zone soil moisture.

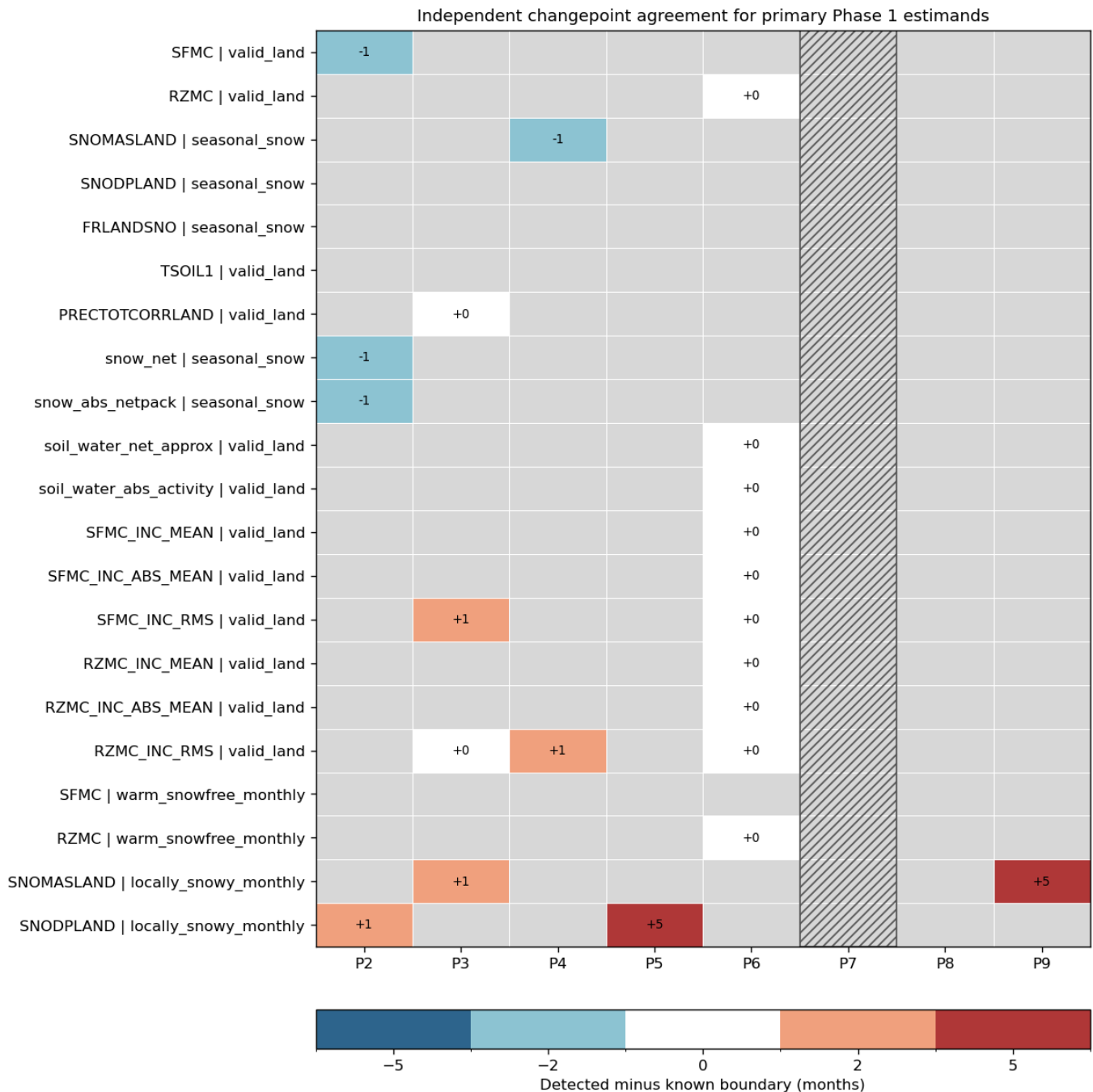
This is a controlled attribution result, not proof that every April 2015 change is caused only by SMAP. The transition date may also coincide with other system or data changes.

Soil-water impacts around the P6 / SMAP transition



Known-date P6 level changes for independently detected April 2015 series





Other observing-system boundaries

- **P2, July 2002 (MODIS Aqua snow cover begins):** signed snow-water correction increases by 2.594 kg m⁻² and absolute snow-correction activity by 4.413 kg m⁻². Both known-date changes survive FDR. Four independent breaks occur within one month of P2.
- **P3, June 2007 (ASCAT-A begins):** SFMC increment RMS has a positive level change. The known-date SFMC DA – 0L slope changes from positive in P2 to negative in P3. Four independent breaks occur within one month, although the independent detector is not reliable for gradual slope hinges.
- **P4, May 2010 (SMOS begins):** two independent snow/root-zone diagnostic breaks occur nearby, but the corresponding known-date coefficients do not survive FDR. These are candidates rather than firm transition findings.

- **P5, P7, P8, and P9:** no primary known-date level or slope change survives FDR. P5 and P9 each have one independent match only under the wider +/-6-month sensitivity. No accepted break matches P7 or P8. Failure to detect a change is not evidence that a sensor had no effect.

Independent Breakpoint Inventory

Across 43 area-weighted monthly series, the conservative consensus accepts 37 breaks: 20 in paired DA – OL series and 17 in DA correction diagnostics. None occurs in the paired OL or DA control series. Twenty accepted breaks match a scored P2-P9 boundary within +/-3 months, and two more match only within +/-6 months. Fifteen are farther than six months from any scored boundary.

Those 15 unmatched breaks are retained as real statistical candidates, but they should not be assigned to the nearest observing-system event without additional metadata or physical evidence. They may reflect climate variability, forcing changes, model behavior, or undocumented system changes.

Interpretation and Limitations

The most scientifically useful result is not that correction activity rises as more observations become available. That behavior is expected. The useful result is that increasing activity generally does **not** produce a widespread long-term drift in the analyzed snow or soil-moisture states.

The conclusions should retain the following limits:

- These are model-internal OL/DA comparisons, not independent validation.
- A whole-record Sen slope can summarize several discrete observing-system steps; the interrupted-series analysis is required for attribution.
- Tile maps establish spatial heterogeneity, while breakpoint results are area-weighted domain means and do not prove spatial uniformity.
- Dynamic warm/snow-free and locally snowy masks change monthly support.
- P7 is only 15 months and cannot support a separate slope claim under the predeclared minimum-segment rule.
- Independent PELT validation is strong for abrupt level shifts but weak for gradual slope changes. A missing independent break does not refute a known-date slope change.
- Spatial significance must be read from `significant_fdr`; approximate Sen confidence intervals are not used to define mapped significance.

Overall Conclusion

Over 2000-2024, OL contains regional precipitation and soil-moisture trend structure and a coherent decline in seasonal-snow-domain snow-covered fraction. DA preserves the precipitation and snow trends, introduces only limited SFMC trend changes, and modifies RZMC trends regionally without creating a significant global land-mean drift. The strongest observing-system event is the April 2015 SMAP transition, which produces an abrupt increase in correction activity and a small positive shift in root-zone DA – OL.

In short:

The evolving observing system changes when and how strongly GEOS LDAS corrects the land state, but it does not impose a widespread artificial long-term trend on snow or soil moisture. Its clearest persistent state effect is regional, positive modification of root-zone soil-moisture trends, with an abrupt system-level shift at SMAP entry.

Reproducibility and Detailed Results

The full contract is in `trends_breakpoints_methods.md`; focused results are in `phase1_trend_results.md`, `phase1_state_trend_results.md`, `phase1_interrupted_series_results.md`, and `phase1_changepoint_results.md`. Figures are generated by `../notebooks/phase1_state_trend_maps.ipynb` and `../notebooks/phase1_trends_breakpoints_summary.ipynb`.

Machine-readable configurations are under `projects/M21C_ls/config/`. Derived NetCDF and CSV products are generated under `projects/M21C_ls/output/trends_breakpoints/` and excluded from version control.